Nuclear Fusion

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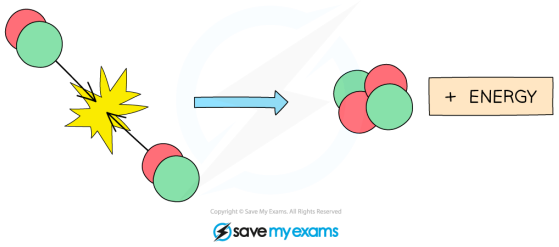
Nuclear fusion

Extended tier only

- The Sun releases energy by the process of nuclear **fusion** in its core
  - Nuclear fusion involves the **fusing** of hydrogen nuclei to form helium nuclei, releasing massive amounts energy in the process

Nuclear fusion



Fusion is the process in which small nuclei, such as hydrogen, are fused together to form larger nuclei, releasing energy in the process

- It is **theoretically possible** to produce a **fusion reactor** that could be used to **generate electricity**
  - This technology could potentially solve the world's energy crisis
- Fusion requires extremely **high temperatures** and **high pressures**, like in the centre of a star
- Scientists are currently researching how to sustain a fusion reaction at lower temperatures
  - International research projects funded by some of the world's largest businesses are making progress with some promising results
- Currently, the fusion reactions require nearly as much energy as they release, but progress is being made toward net energy production
  - If they succeed, virtually limitless amounts of energy could be produced, with large-scale, carbon-free electricity generation

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